SPINOTOR – Auto-Calibration Framework

Technical Reference Manual

Theory, Methods and Implementation of Automated Parameter Identification

Document Information

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Abstract

This document presents the complete auto-calibration framework of the SPINOTOR motor drive simulation platform. For each subsystem subject to automatic parameter identification—PI current and speed regulators, PLL observer, SMO, ST-SMO, Active Flux observer, SOGI adaptive filter, Extended EMF (EEMF) model, field-weakening controller, startup sequencer, and Power Factor Controller (PFC)—the report provides: (i) *the theoretical foundation from classical control and electrical machine theory,* (ii) *the calibration methodology with analytical formulae and algorithmic description, and* (iii) *illustrative figures generated directly by the simulator. The exposition follows IEEE journal conventions with full bibliographic references.*

Index Terms**—***SPINOTOR, BLDC motor, PMSM, auto-calibration, FOC, sliding mode observer, PLL, active flux, SOGI, field weakening, sensorless control, PI tuning.*

I. Introduction

Modern BLDC and PMSM drives require precise tuning of multiple control loops and estimation algorithms to achieve high dynamic performance, energy efficiency, and safe operation over a wide speed range [1, 2]. Manual parameter selection is error-prone and motor-dependent [3]. The SPINOTOR platform integrates a comprehensive auto-calibration pipeline that derives controller gains and observer parameters analytically from motor nameplate data, then validates them through frequency-domain analysis and closed-loop simulation [4].

Stage 1 determines the PI gains for the d-q current and speed loops using bandwidth-based analytical design reinforced by a multi-resolution grid search [6]. **Stage 2** handles field-weakening calibration through physics-based computation followed by a simulation sweep [5].

II. PI Current and Speed Regulators

A. Theory

The d-q current dynamics of a PMSM in the rotating frame are described by the first-order transfer function [2, 6]:

$$G_i(s) = \frac{1}{Ls + R} \tag{1}$$

A PI controller $C_i(s) = K_p + K_i/s$ forms a type-1 closed loop with zero steady-state error [3]. The IMC bandwidth design [4, 7] gives the analytical gains:

$$K_p = \omega_c L, \qquad K_i = \omega_c R \tag{2}$$

For the outer speed loop [9]:

$$K_{p,\omega} = \frac{\omega_{c,\omega} J}{K_t}, \quad K_{i,\omega} = \frac{\omega_{c,\omega} B}{K_t} \tag{3}$$

Anti-windup via back-calculation [3]:

$$\frac{dI}{dt} = e + K_{\text{aw}}(u_{\text{sat}} - u_{\text{unsat}}) \tag{4}$$

B. Calibration Methodology

Step 1—Analytical guess: K_p, K_i computed from (R, L, K_t, J, B) via bandwidth formulae [4].

Step 2—Coarse grid: 8×8 log grid, ±50% around the analytical guess; margins evaluated over 2000 frequencies; controllability and observability checked by Kalman rank condition [8].

Step 3—Fine grid: 10×10 grid, ±25% around best coarse candidate. Targets: GM≥ 6 dB, PM≥ 45° [3, 9].

C. D/Q Feed-Forward Decoupling

Feed-forward voltages $v_{d,\text{ff}} = -\omega_e L_q i_q$ and $v_{q,\text{ff}} = \omega_e(L_d i_d + \lambda_{\text{pm}})$ are added to the PI outputs, rendering both axes independently regulated [2, 6]. See Section 12.

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III. PLL Observer

A. Theory

The PLL estimates $\hat{\theta}_e$ and $\hat{\omega}_e$ by tracking the reconstructed back-EMF phase angle [10,11]:

$$e_\alpha = v_\alpha - Ri_\alpha - L\,\frac{di_\alpha}{dt} \tag{5}$$

$$e_\beta = v_\beta - Ri_\beta - L\,\frac{di_\beta}{dt} \tag{6}$$

Second-order PLL parameterization [12]:

$$K_p = 2\zeta\omega_n, \qquad K_i = \omega_n^2 \tag{7}$$

B. Calibration Methodology

$$\omega_n = \mathrm{clip}(0.10\,\omega_{e,\max}, 50, 600)\ \mathrm{rad/s} \tag{8}$$

Damping ratio $\zeta = 0.9$ provides slightly over-damped convergence [13]. When adaptive PLL bandwidth is active, $\omega_n \propto |\hat{\omega}_e|$ [13]. Recommended for isotropic motors with $\omega_{e,\max} > 1500\,\mathrm{rad/s}$ (Section 13) [41,43].

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IV. Sliding Mode Observer (SMO)

A. Theory

The SMO uses the $\alpha\beta$-frame motor model [14, 15]:

$$L\,\frac{d\hat{i}_\alpha}{dt} = v_\alpha - R\,\hat{i}_\alpha - z_\alpha \tag{9}$$

$$z_\alpha = K_{\text{slide}}\ \text{sat}(\hat{i}_\alpha - i_\alpha, \Delta) \tag{10}$$

In sliding mode, the equivalent control converges to the back-EMF, extracted by a LPF [14, 16]:

$$\hat{e}_\alpha[k+1] = (1-\alpha)\hat{e}_\alpha[k] + \alpha z_\alpha[k] \tag{11}$$

$\hat{\theta}_e = \text{atan2}(\hat{e}_\beta, \hat{e}_\alpha) - \pi/2$.

B. Calibration Methodology

Reachability condition [14, 15]:

$$K_{\text{slide}} = \max(50, 1.5\,K_e\,\omega_{e,\max}) \times 1.5 \tag{12}$$

$\tau_{\text{LPF}} = 3L/R$; boundary layer $\Delta = 0.06\,K_{\text{slide}}/100$. For salient IPM motors ($L_d \neq L_q$) the missing term $(L_q - L_d)i_d\omega_e$ biases the SMO; the Active Flux observer must be used instead [17, 22].

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V. Super-Twisting SMO (ST-SMO)

A. Theory

The Super-Twisting algorithm [18] converges to the back-EMF in finite time without a LPF [19]:

$$z_\alpha^{(1)} = K_1 |\sigma_\alpha|^{1/2} \operatorname{sign}(\sigma_\alpha) + z_\alpha^{(2)} \tag{13}$$

$$\frac{dz_\alpha^{(2)}}{dt} = K_2 \operatorname{sign}(\sigma_\alpha) \tag{14}$$

At steady state $z_\alpha^{(2)} \to \hat{e}_\alpha$: zero phase lag [20].

B. Calibration Methodology

$$K_1 = \lambda K_e \omega_{e,\max} \cdot 2.0, \quad K_2 = \max(50, 1.5 K_e \omega_{e,\max}) \cdot 1.5 \tag{15}$$

$\lambda = 3.0$ (isotropic), $\lambda = 4.0$ (salient motors [17, 21]). Finite-time convergence condition $K_1 > 2\sqrt{\Gamma_M/\Gamma_m}$ verified for all accepted motors [18].

VI. Active Flux Observer

A. Theory

The Active Flux concept [22] defines $\boldsymbol{\psi}_a = \boldsymbol{\psi}_s - L_d \mathbf{i}_s$. For an IPM motor, $\boldsymbol{\psi}_a$ always points along the d-axis [23]:

$$\hat{\theta}_e = \operatorname{atan2}(\psi_{a,\beta}, \psi_{a,\alpha}) \tag{16}$$

Stator flux estimated by voltage-model integration with drift correction [24, 25]:

$$\frac{d\psi}{dt}_{s,\alpha} = v_\alpha - R\, i_\alpha \tag{17}$$

B. Calibration Methodology

Drift corrector cutoff frequency [22, 24]:

$$f_c = \operatorname{clip}\left(\frac{f_{e,\min}}{10}, 0.05, 0.5\right) \text{ Hz} \tag{18}$$

Automatically selected for salient IPM motors ($L_q/L_d > 1.2$) [22, 25].

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VII. Extended EMF (EEMF) Model

A. Theory

For salient motors, the standard EMF model introduces an angle error $\propto (L_q - L_d) i_d \omega_e$ [26]. The EEMF model [26] substitutes L_q in the di/dt term:

$$e_{\mathrm{EEMF},\alpha} = v_\alpha - R i_\alpha - L_q \frac{di_\alpha}{dt} \tag{19}$$

$$e_{\mathrm{EEMF},\beta} = v_\beta - R i_\beta - L_q \frac{di_\beta}{dt} \tag{20}$$

The resulting EEMF vector is proportional to $[-\sin\theta_e; \cos\theta_e]$, enabling unbiased position estimation [26, 27].

B. Calibration Methodology

If $\sigma = L_q/L_d > 1.2$, the EEMF model is activated together with the SOGI filter (Section 8) [26,28]. The MRAS resistance adaptation gain is [28]:

$$\gamma_R = \frac{1}{0.5\, R\, I_{\max}^2} \tag{21}$$

VIII. SOGI Adaptive EMF Filter

A. Theory

The SOGI [29,30] replaces the fixed-frequency LPF with a speed-adaptive resonant bandpass:

$$H_{\mathrm{SOGI}}(s) = \frac{k\,\omega_e\,s}{s^2 + k\,\omega_e\,s + \omega_e^2}, \quad k = \sqrt{2} \tag{22}$$

Key property: $\angle H_{\mathrm{SOGI}}(j\omega_e) = 0^\circ$ – zero phase lag at resonance, eliminating the LPF-induced position error [29,31].

B. Calibration Methodology

No gain tuning required: $k = \sqrt{2}$ is fixed by critical damping [29]. ω_e updated adaptively at each step [30]. Automatically activated with the EEMF model or ST-SMO mode.

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IX. Field-Weakening Auto-Calibration

A. Theory

Above N_{base}, the back-EMF exceeds $V_{\text{ph}} = V_{\text{dc}}/\sqrt{3}$. Negative I_d reduces the effective PM flux linkage [32, 33]:

$$\lambda_{\text{pm,eff}} = \lambda_{\text{pm}} \cdot \text{clip}(1 + k_{\text{fw}} \min(I_d, 0), r_{\min}, 1) \tag{23}$$

Minimum flux ratio to reach rated speed [33]:

$$r_{\text{req}} = \frac{V_{\text{ph}}}{K_e\, \omega_{m,\text{rated}}} \tag{24}$$

FOC voltage headroom integrator [34]:

$$\frac{dI_{d,\text{fw}}}{dt} = K_{\text{fw}}(V_{\text{hdm}} - (V_{\text{lim}} - |V_{dq}|)) \tag{25}$$

B. Calibration Methodology

Stage 1 (analytical):

$$k_{\text{fw}} = \frac{(1 - r_{\text{req}}) \cdot 1.05}{I_{d,\max}},\ I_{d,\max} = 0.7\, I_{\text{rated}} \tag{26}$$

$V_{\text{hdm}} = 0.05\, V_{\text{ph}}$ [32, 33].

Stage 2 (simulation sweep): 3×3 gain grid $\{0.5, 1.0, 1.5\} \times K_{\text{fw,analytical}}$ over three operating points (50%, 75%, 100% load) [5, 34].

X. Startup Sequence Auto-Tuning

A. Theory

At standstill the back-EMF is zero and observer convergence requires a minimum speed [36,37]. Three phases are implemented.

Phase 1—Alignment: Constant I_d at θ=0, forcing the rotor to align with the reference frame [35, 36].

Phase 2—Open-loop ramp: Reference angle incremented at a controlled rate (V/Hz mode) [36, 37].

Phase 3—Observer handoff: Switch to observer estimate when confidence metric C exceeds the hysteresis band [37, 41]:

$$C = w_{\text{emf}}\, C_{\text{emf}} + w_{\omega}\, C_{\omega} + w_{\text{coh}}\, C_{\text{coh}} \tag{27}$$

B. Calibration Methodology

$$t_{\text{align}} = 5\tau_e = 5L/R \tag{28}$$

$$I_{\text{align}} = \text{clip}(0.10\, V_{\text{bus}}/R, 0, I_{\text{rated}}) \tag{29}$$

$$I_{q,\text{startup}} = 0.05\, V_{\text{bus}}/R \tag{30}$$

where t_{align} gives 99.3% settling [35] and $I_{q,\text{startup}}$ follows [36, 37].

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XI. Power Factor Controller (PFC)

A. Theory

The PFC compensates reactive power to approach unity input power factor [38, 39]. Over a sliding window of N_w samples:

$$P = \frac{1}{N_w} \sum_k v_k i_k, \quad S = V_{\text{rms}} I_{\text{rms}}, \quad Q = \sqrt{S^2 - P^2} \tag{31}$$

B. Calibration Methodology

Maximum compensation variance [38]:

$$\text{max_var} = \text{clip}\left(\frac{V_{\text{bus}}^2}{2R} \cdot 0.30, 100, 50000 \right) \tag{32}$$

Window length covering at least two electrical periods [39]:

$$N_w = \text{clip}\left(\frac{2}{f_{e,\text{rated}}\, dt}, 8, 2000 \right) \tag{33}$$

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XII. D/Q Axis Decoupling

A. Theory

Cross-coupling terms in the dq voltage equations [2, 40]:

$$v_d = Ri_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \tag{34}$$

$$v_q = Ri_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \lambda_{\mathrm{pm}}) \tag{35}$$

Feed-forward decoupling [2, 6, 40]:

$$v_d = v_{d,\mathrm{PI}} - \omega_e L_q i_q \tag{36}$$

$$v_q = v_{q,\mathrm{PI}} + \omega_e (L_d i_d + \lambda_{\mathrm{pm}}) \tag{37}$$

B. Calibration Methodology

Always enabled post-calibration. Gains computed from L_q, L_d, and $\lambda_{\mathrm{pm}} = K_e/p_p$ [2, 6, 40].

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XIII. Observer Auto-Selection

A. Selection Criteria

Criterion 1—Saliency: For $\sigma = L_q/L_d > 1.2$ (IPM), standard back-EMF observers suffer a bias $\propto (L_q - L_d) i_d \omega_e$ [17, 26]. The Active Flux observer is immune [22, 25].

Criterion 2—Maximum electrical speed: Below $\omega_{e,\max} \approx 1500\,\mathrm{rad/s}$, the SMO is more robust at low SNR. Above this threshold, the SMO LPF phase lag becomes excessive and the PLL is superior [41–43].

B. Decision Tree

Branch 1 ($\sigma > 1.2 \Rightarrow$ ActiveFlux) [22].
Branch 2 (isotropic, $\omega_{e,\max} > 1500 \Rightarrow$ PLL) [10, 13].
Branch 3 (isotropic, $\omega_{e,\max} \leq 1500 \Rightarrow$ SMO) [14, 41].

C. Non-Regression Validation

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XIV. Calibration Pipeline Summary

A. Execution Order

1. **Stage 1:** `tune_analytical()` – PI gains, observer parameters, D/Q decoupling, startup timing [4, 6].
2. **Stage 2:** `calibrate()` – analytical FW parameters and simulation sweep [5, 34].
3. **Observer selection:** `_recommend_observer()` [22, 41].
4. **GUI update:** All calibrated parameters applied.

B. Motor Fingerprint Cache

Results cached by fingerprint $(R, L, K_e, V_{\text{nom}}, p_p)$ with 3% tolerance [5].

C. Summary Table

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